Theoretical Determination of the Viability of Quantum Cryptography

RAY, DHEEMAAN[1] and YUEN, JACKSON[1]

[1]Department of Physics, Syracuse University, Syracuse, NY 13210, USA

ABSTRACT.
The purpose of this research is to demonstrate how quantum concepts relating to the state of photons as quantum particles can be used to validate the BB84 cryptography protocol to encrypt communications – the use of single photons can be used to encode information while their polarizations encrypt that information. To achieve this demonstration, a statistical analysis of bit-match possibilities in key generations is used to theoretically demonstrate the workings of the BB84 protocol; while quantum concepts like Heisenberg's Uncertainty Principle and Wootter's "No-Cloning" theorem, which constitute the foundation of those workings, is also discussed. Physically, ThorLabs Quantum Cryptography Kit is used to pulse light, polarize it, and detect it via a beamsplitter to tangibly demonstrate the viability of the BB84 protocol. In doing so, the theoretical expectation that an eavesdropper can be detected 100% of the time is achieved by demonstrating that the error in bit-matching approaches 25% if an eavesdropper is present.

I. INTRODUCTION

The significance of this experiment is to demonstrate the feasibility of the application of quantum concepts to create an infallible and tamper-proof communication encryption technique by enabling the detection of an eavesdropper with 100% accuracy. As such, this manuscript demonstrates how Quantum Key Distribution (QKD) enables secure communication by leveraging quantum mechanics to detect eavesdropping. The BB84 Protocol, central to this work, uses photon polarization to encode key-bits, ensuring security through the No-Cloning Theorem, Heisenberg's Uncertainty Principle, and resultant measurement-induced disturbances. The use of such a technique has significant implications in personal privacy, business, and national security from the perspective of preventing unwanted monitoring of data exchanges.

A. Background

Classical cryptography as a field can generally be described as a method to store and transmit data in such a way that the data can only be read by who it was intended for. The origins of the field can be traced all the way back to Ancient Egypt in 1900 BC, while modern cryptography arose as a necessity to encrypt communications in World War I and II[1].

Modern cryptography falls under two categories: Secret Key and Public Key cryptography. Public Key cryptography classically uses a public key for data to be encrypted with a private key, and the data encrypted with a private key can only be read with a public key. Quantum Key Distribution, or QKD is the implementation of Public Key cryptography, and the BB84 Protocol is an application of Public QKD. The BB84 Protocol itself is an encryption protocol developed by Charles Bennett and Gilles Brassard in 1984 and enables the users to predict the existence of an eavesdropper 100% of the time, theoretically[2]. It uses Heisenberg's Uncertainty Principle and Wootter's "No-Cloning" Theorem, thereby applying quantum principles to create infallible encryption for communication.

B. Theory

1. The One-Time Pad

The point of this protocol is to be able to develop a key, publicly, between sender (Alice) and receiver (Bob), through which to encrypt future communications. As such, the foundation of BB84 cryptography is the one-time pad: bits of 1s and 0s can form an encryption key, because when added to data in the form of 1s and 0s, the resulting transmission is essentially gibberish which cannot be read without the key[3].

TABLE I: Example of the One-Time Pad

Data:	0	1	1
Encryption Key	1	0	0
Encrypted Data	1	1	1

In BB84, these bits are encoded into the polarizations of single photons.

2. Polarization, Basis, and Bits

The big issue is that in developing the key itself, an eavesdropper (Eve) could corrupt the encryption by either (a) copying the key in transit from Alice to Bob or (b) reading the key sent by Alice and retransmitting it to Bob. Since this method utilizes single photons, Wootter's "No-Cloning" theorem comes into play – since BB84 encodes key bits into the polarization states of single photons, any attempt at "cloning" the photon requires measuring it. The measurement itself inevitably disturbs the quantum state of the photon, and this state cannot then be duplicated in another measurement[4] due to fundamental quantum principles limited by Heisenberg's Uncertainty Principle, by

which it is impossible to simultaneously measure conjugate quantum states[5]. This means that Eve cannot copy the key in transit from Alice to Bob; she only has one option – to read the key and retransmit it to Bob.

Since light is an electromagnetic wave with oscillating electric and magnetic fields perpendicular to each other and the direction of propagation, the polarization of photons refers to the orientation of the electric vector of a light wave as it propagates[6]. This fact is exploited by BB84, as the polarization of light is used to encode bits of 0s and 1s. This is done by first defining bases of "+" and "x", which stand for rectilinear and diagonal polarizations of light, and define the conjugate states to be framed by Heisenberg's Uncertainty Principle. This definition is as follows:

TABLE II: Polarization Definition of "Basis"

	"0" Bit	"1" Bit
"+" Basis	0°	90°
"x" Basis	- 45°	45°

For example, to encode "0 1 0" in a "+ x x" basis, three successive photons would be polarized as 0°, -45°, and 45°. This is done using polarizers; then a beamsplitter on the receiver's end directs the photon into a detector for 0°, labeled bit "0", and 90°, labeled bit "1". As will be discussed in the next section, there is also a possibility the polarization of the photon received can be -45°, 45°, and 135°. What happens then? Theoretically, both sensors should light up, and 0 or 1 is chosen randomly[7].

3. The QKD Ensemble

We have now established that in communication between Alice and Bob, if an eavesdropper Eve exists, Eve is forced to intercept information from Alice and retransmit it to Bob in the hopes of learning what the key the two of them develop is to decrypt future communications. Let's approach the detection of Eve first without her presence, and then with. In either case, the steps taken to ensure encryption are thus:

1) Alice sends "test" bits to Bob

Alice chooses bits and bases *randomly*. Bob chooses a basis to receive the bits *randomly* as well.

2) Comparison after each bit

After every bit, Alice and Bob compare *bases*. If Bob chose wrong and the bases don't match, Bob *discards* the measurement. This means that *50% of test bits are actually measured* (as the number of test bits approaches infinity).

3) Alice and Bob Compare Bits

At this point, if Eve doesn't exist, then all bits should match exactly. These bits are then used to construct the key used for communication encryption.

TABLE III: Encryption Key Generation without Eavesdropper

Alice	+	x	x	x	+	x	+
	1	1	0	1	0	0	0
Eve	x	x	+	x	+	+	+
	0	1	0	1	0	1	0
Bob	+	+	x	x	+	+	x
	0	*Disc.*	0	1	0	*Disc.*	*Disc.*
Key	**0 0 1 0**						

Now if Eve is present, then she must *also guess* the basis in which Alice has sent a test bit, same as Bob. There are then only 4 possibilities:

1) Eve guesses *correctly* | Bob guesses *correctly* = bit is *kept and accurate*

2) Eve guesses *correctly* | Bob guesses *incorrectly* = bit is *discarded*

3) Eve guesses incorrectly | Bob guesses incorrectly = bit is discarded

4) Eve guesses *INCORRECTLY* but Bob guesses *CORRECTLY* = bit is *KEPT* and potentially *INACCURATE*)
Eve guesses *INCORRECTLY* | Bob guesses *CORRECTLY* = bit is *KEPT* and potentially *INACCURATE*

This last possibility results in a situation where the addition of polarizations comes out to -45°, 45°, and 135°.

TABLE IV: Aberrant Basis Combinations

Basis Combo	"0" Bit	"1" Bit
"+" → "x"	0°+(-45°) = -45° ~ ***?***	90° + 45° = 135° ~ ***?***
"x" → "+"	-45°+ 0° = -45° ~ ***?***	45° + 0° = 45° ~ ***?***

This means that as mentioned before, 0 or 1 is chosen *randomly*. Since both Eve and Bob must randomly choose 0 or 1 in this scenario at a 50% accuracy rate, then the ***error in test bits*** in the key construction approaches ***25%*** with infinite test bits. Similarly, without Eve, the error in test bits approaches ***0%***.

TABLE V: Encryption Key Generation with Eavesdropper

Alice	+	x	x	x	+	x	+
	1	1	0	1	0	0	0
Eve	x	x	+	x	+	+	+
	0	1	0	1	0	1	0
Bob	+	+	x	x	+	+	x
	0	*Disc.*	0	1	0	*Disc.*	*Disc.*
Key	**0 0 1 0**						

II. METHODS

A. Materials and Equipment

The equipment used in this experiment can generally be categorized into equipment used by the sender (Alice) and receiver (Bob). The equipment used by the eavesdropper (Eve) includes both sender and receiver components. A general schematic of the setup can be seen in Figure 1. All equipment units are attached to their respective base plates via screws and are held in adjustable stands placed at equidistance from each other.

1. Laser

Ideally, quantum cryptography would only use single photons to transmit information, ensuring that they cannot be copied or measured. However, this experiment uses a classic light source in the form of a laser. The laser is used by the sender to emit photons to the receiver; en route, the photons will be polarized by both the sender and receiver and the basis and bits will be recorded by each party. To limit the amount of photons emitted, the laser is first weakened by decreasing the intensity of the light[2]. The photons are then sent in pulses by pressing the fire button on the laser electronics control.

2. Polarizing Rotator Plate

For this experiment, the photons being sent and received will be polarized using λ/2 polarization rotator plates[8]. These plates consist of a liquid crystal polymer in between two glass plates. Optical wave plates of this nature use birefringent materials that split light due to a difference in refractive index between two orthogonal axes, known as the fast and slow axes[9]. These optical axes are parallel to the surface of the plate and result from the orientation of the cut of the crystal and thickness. Specifically, the fast axis has a lower refractive index than the slow axis such that when a light wave is normally incident on the plate, the wave velocity of the light polarized in this direction will be higher than that of the slow axis. This velocity difference results in a phase shift that determines the polarization state of the light exiting the plate.

This type of plate linearly rotates the polarization of the incoming photons by double the physical angle that the plate itself is rotated; for example, rotating the plate itself by 45° relative to incoming polarization results in a 90° polarization of the photon. In order for the use of two bits (0 and 1) and two bases (x and +), the sender will require a polarizing rotator capable of four different angles of polarization, -45°, 0°, 45°, 90°; the receiver will only need a 0°, 45° polarizing rotator to translate the photons into an x or + basis[2]. Like for the laser, the λ/2 plates are first oriented and reconfigured to minimize the reflected intensity of the laser.

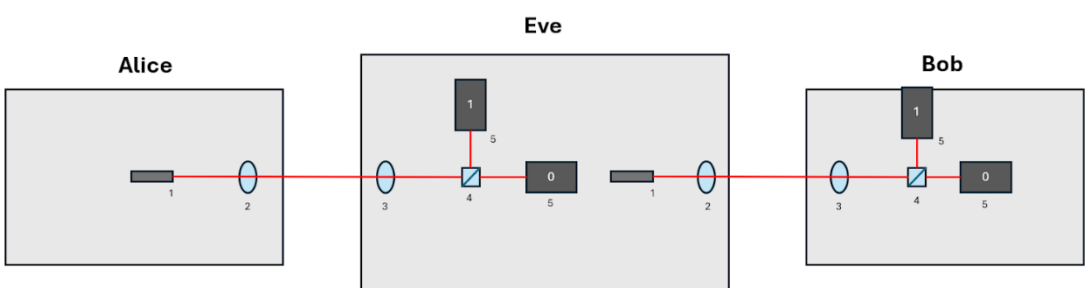

Diagram I: schematic representation of the experiment set up. All components are held up on bases and attached to the base plate with screws.

3. Polarizing Beamsplitter Cube

After the receiver has chosen which base to receive the photons and altered the polarization plate accordingly, the light is passed through to a polarizing beamsplitter cube. This cube consists of two N-SF1 glass prism halves glued together with cement[10]. This orientation causes incident light to be split into reflected s-polarized (vertical 90º component) and transmitted p-polarized (horizontal 0º component) beams, which are then received by the sensor modules[11]. For ±45° polarized light, the photons are in an equal superposition of horizontal (H) and vertical (V) polarization states. When they encounter a polarizing surface, such as a beamsplitter, the surface measures the photons in a H/V basis, causing its polarization to collapse into either H or V with equal probability. Since the beamsplitter transmits H components and reflects V components, an incident ±45° polarized photon will be transmitted 50% of the time and reflected 50% of the time[2]. Experimentally, this means that the light will be simultaneously reflected and transmitted if polarized in a ±45° basis.

4. Light Sensor

After passing through the beamsplitter cube, the light will then be received by a light sensor. The light sensor detects whether the photons are reflected, transmitted, or both. If photons are detected, an LED on the top of the sensor lights up as an indicator. If light is transmitted, the corresponding bit that is recorded by the receiver is a "0"; if it is reflected, it is a "1". In the event that the light is both transmitted and reflected, a "0" or "1" bit is chosen at random.

B. Procedure

After the equipment has been set up and verified, the random bases (x, +) for Alice, Bob, and Eve and the random bits (0, 1) for Alice need to be generated. The length of the bases and bits can be determined at random or depending on the size of a message. For the latter, the length is determined by doubling the size of the message in binary, as each letter of the alphabet has a 7-digit binary representation (for example, the word "test" has 4 letters and will therefore have a binary length of 28 and will need a test bit length of 56). Once the length of the bases and bits has been determined, the actual test bits and bases can be generated randomly, either through dice rolls, coin flips, or random number generators (RNG). Generally, at test bit size larger than 50 is needed to reliably detect an eavesdropper. The sender than sets their polarization depending on the basis they want to transmit their bit in according to Table II. The eavesdropper also sets their polarization to either 0º or 45º for an + or x basis respectively. The laser is then pulsed and the eavesdropper records which bit is received (if the light is both transmitted and reflected, a "0" or "1" is randomly chosen). The eavesdropper then uses their same chosen basis to transmit the bit they just received to the receiver, again using Table II. The receiver also sets their polarization 0º or 45º depending on their random basis. The eavesdropper then sends the photons to the receiver, and the receiver records the bit.

C. Analysis

After the receiver records each bit, they compare their bases with the sender. If the bases match, the bases and bits are kept and stored as part of the generated encryption key; otherwise, the measurement is discarded. The sender and receiver then compare their respective encryption keys. All of the bases in the key should match; of any of the bits do not match, an eavesdropper is detected. A simple example of the series of measurements and analysis to detect an eavesdropper can be seen in Table V. The difference in bits between the sender and receiver can be quantified using a simple percent error calculation:

$$PE = \frac{L-K}{L}(100\%) \qquad [1]$$

where L is the length of the generated encryption keys and K is the number of bits of the encryption key that match between the sender and receiver. As the size of the test bits approaches infinity, the percent error in the generated keys should approach 25%; regardless, however, any percent error in the generated keys implies the presence of an eavesdropper.

D. Simulation

As well as being verified with the equipment, the ability to generate encrypted keys and detect eavesdroppers was also verified symbolically through Python simulations. The benefit of simulating this process is that it allows for much larger test bit sizes and the ability to see how the percent error in key generation changes across various test bit sizes. The program also allows for more efficient trials, as it can generate test bit sizes and carry out the process extremely quickly relative to how fast it can be done by using the equipment.

The Python program that was used follows the same overall procedure. The program first generates test bits and bases depending on either the size of an input message or an arbitrarily set size. Polarizations are then set for the sender, eavesdropper, and receiver symbolically based on the test bits and bases and the corresponding bit is generated for the receiver. The percent error in key generation due to the

eavesdropper is then calculated in the same way and reported.

This general program was also used to view the characteristics of the percent error in key generation over various test bit sizes. For this, various test bit sizes (10, 50, 100, 500, 1,000, 5,000, 10,000) are chosen and the same overall procedure is followed. For each test bit size, 1000 individual simulations are run. The results can then be collectively plotted and examined statistically to analyze certain trends based on test bit size.

The Python program that simulates individual, general trials and the program that simulates multiples trials based on test bit size with statistical analysis included separately as Annex files 1 and 2 respectively with this manuscript.

III. UNCERTAINTY

Based on the nature of this experiment, the statistical analysis surrounding the uncertainty that can be conducted is somewhat limited. The bits and bases used by Alice, the bases used by Eve, and the bases used by Bob will all be randomly generated either using dice or a random number generator (RNG) where there should always be a 50% of choosing a "x" or "+" basis or "0" or "1" test bit, especially over a sufficiently large set. The recommended size of a set for test bits is 52 bits long. This experiment essentially relies on determining whether or not any error in test bits between Alice and Bob is observed after they compare the bits in their key, corresponding to the presence of an eavesdropper, Eve, with the error approaching 25% with infinite measurements test bits. This general principle is demonstrated using the Python program to simulate 1000 tests of varying test bit sizes and can be seen in Figure 1. From here, further analysis can be done to determine the standard deviation in percent error in key generation per test bit size, seen in Figure 2. This shows the overall dispersion of the percent errors in key generation per test bit size around the mean, calculated from Equation [2]. The standard error of the mean (SEM) is also shown in Figure 3 as a measurement of the variability of the sample mean estimate from the true population mean. It indicates how much the sample mean is likely to differ from the actual population mean, with a smaller SEM suggesting more precise results. The SEM is calculated from Equation [3]. Finally, the standard deviation of the mean around 25% can be seen in Figure 4. This measurement represents how much the mean values of the percent error per test bit size vary around the 25% threshold and is another indicator of the accuracy of the results based on test bit size. The standard deviation of the mean around 25% is calculated from Equation [4].

While each measurement of uncertainty provides a different view of the overall accuracy of each test bit size, the characteristics are mostly uniform. All figures show clear trends of increases in accuracy as test bit size is increased, aligning with expectations that larger test bit sizes yield more accurate and reliable results.

One further source of uncertainty in this experiment was caused by an error in the equipment. When the light was polarized at ±45° and should have been both reflected and transmitted, resulting in a random "0" or "1" bit, there were instead distinct outcomes. In this case, the light was either reflected or transmitted instead of both, with the exact equipment response given in Table IV in the Appendix. While the exact source of this error is not known, given the nature of the equipment response, there was still technically a symbolic 50% chance of recording a "0" or "1" bit, especially since the equipment error was consistent. As such, this equipment error proved to be negligible in producing results. However, it is still accounted for in the Python program. All relevant figures subsequently show the uncertainty measurements for perfect symbolic trials as well as for trials with the equipment error. As expected, the nature of the error makes it so that it has no actual impact on the results, as the results are roughly the same both with and without equipment error present.

IV. RESULTS AND DISCUSSION

Using the equipment and a 52-test bit size yielded a roughly 17% error in key generation between the sender and the receiver, corresponding to an accurate detection of the eavesdropper. For simulations carried out with the Python programs, the eavesdropper was detected in the vast majority of trials, as can be seen in Figure 1. However, there was at least one instance with a test bit size of 10 where the eavesdropper remained undetected, as there was no difference in the bits of the generated keys of the sender and the receiver. This inability to accurately detect the eavesdropper is a result of the small size of the test bit. While it is not guaranteed that a test bit size of 10 will fail to detect the eavesdropper, the small sample size means that the probability that the eavesdropper is able to randomly match the bits that the sender generates is relatively large. Given that 1000 simulations that were done per test bit size, a case of zero percent error in key generation was bound to be present with this test bit

Figure 1

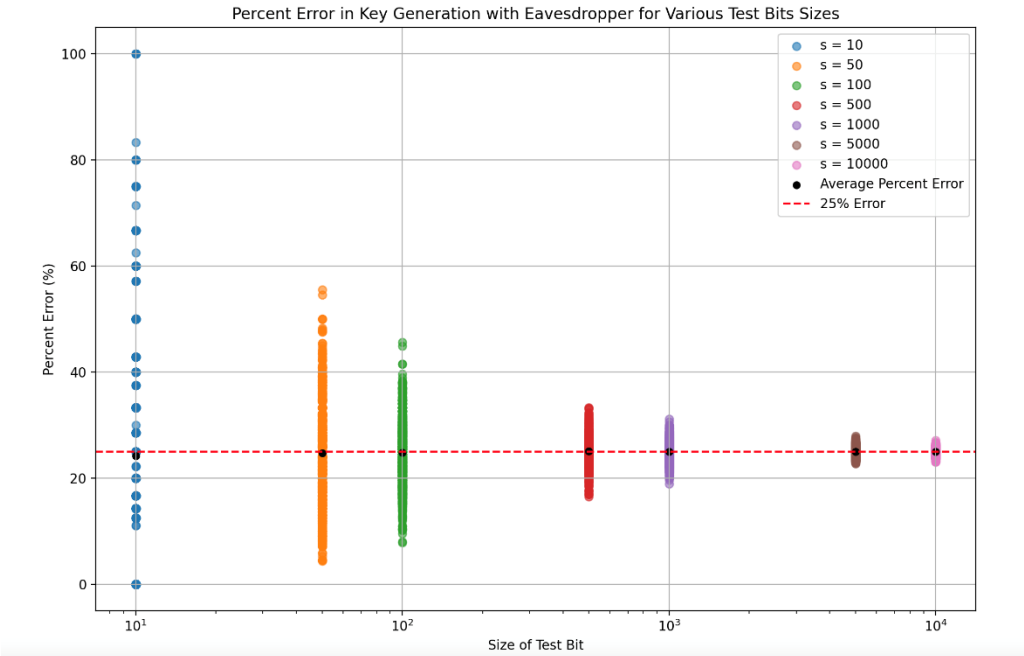

Figure 1: plot of percent error in key generation per test bit size for various test bit sizes

Figure 2

Figure 2: plot of the standard deviation in percent error per test bit size

Figure 3

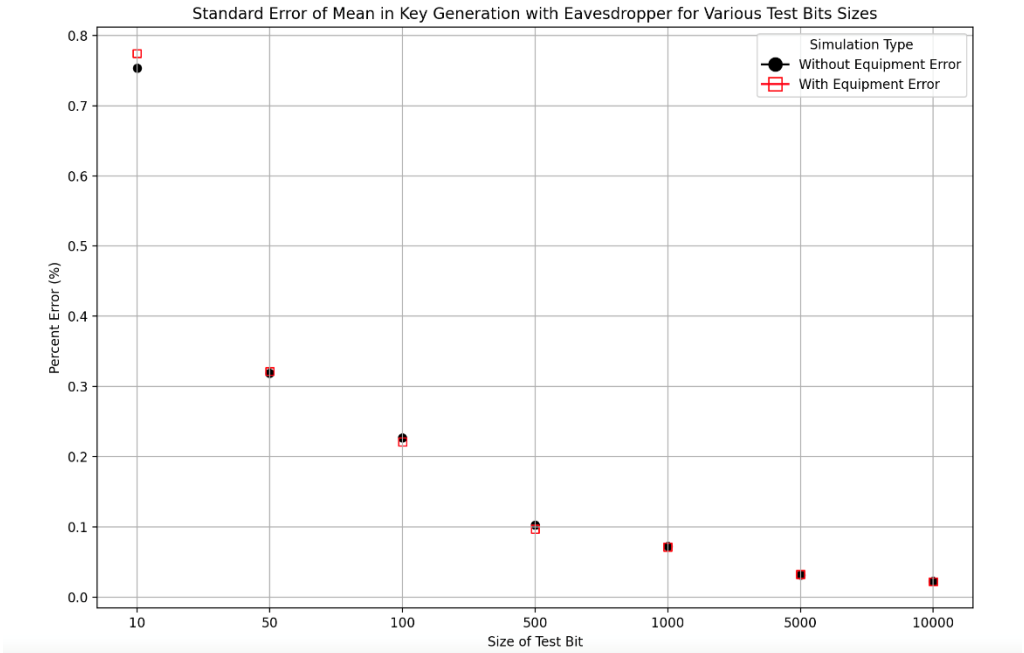

Figure 3: plot of the standard error of the mean in percent error per test bit size

Figure 4

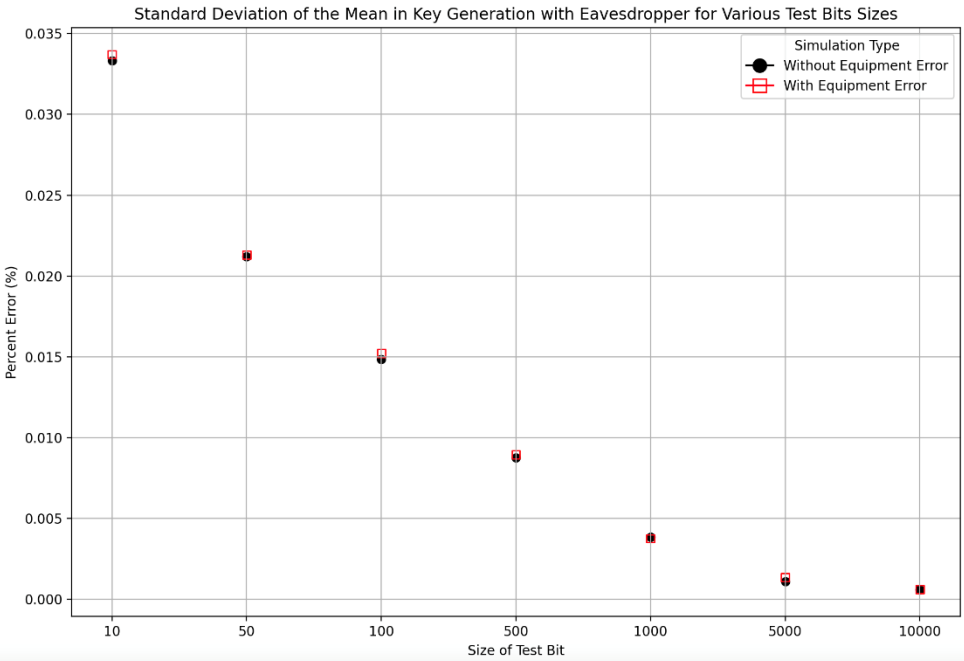

Figure 4: standard deviation of the mean around 25% in percent error per test bit size

size. For all other test bit sizes greater than 10 simulated, the eavesdropper was always detected.

Figure 1 also demonstrates the principle that that the percent error in key generation will always approach 25% as the test bit size approaches infinity. It also demonstrates, on a surface level, that the overall spread of the percent error for any test bit size becomes much more uniform around 25% as the test bit size approaches infinity. Both of these principles align with theoretical expectations. This is generally caused by the larger degree of uniformity that is present with higher test bit sizes compared to smaller test bit sizes, which are much more sensitive to random events and generation.

The statistical analysis conducted also meets expectations. Figures 2, 3, and 4 further demonstrate that larger test bit sizes will lead to a less variance in percent error in key generation as well as higher accuracy around 25%. They also show that the overall nature of the equipment error essentially renders it negligible for quantitative results, as the results are roughly the same regardless of whether the error is present or not. While these figures shows the consistent characterization of the data given the size of the test bit, one possible avenue of further analysis could be to determine exact probability that the eavesdropper will not be detected depending on the size of the test bit. This can likely be determined empirically through slight modifications to the existing programs, though it is also possible that, given the logical nature of the process, a governing equation can be derived that can express this probability. Further testing can also be done to determine the root cause of the error in the random events and whether it is due to a malfunction in the light sensors, a malfunction in the sensor connection unit, or if there is another issue.

V. CONCLUSIONS

This experiment confirms the efficacy of quantum cryptography as an encryption mechanism to generate secure keys and to detect eavesdropper. However, it also emphasizes the necessity of large test bit sizes to accurately detect an eavesdropper and the benefits that it brings around the accuracy of the results.

One important consideration is that while this experiment confirms the theoretical viability of quantum cryptography, both in terms of the physical and cryptographic mechanism, the absence of the transmission of single photons invalidates this as a true means of quantum cryptography. While a laser can be pulsed to limit the amount of photons emitted, the presence of a photon stream nullifies the security of from the non-cloning and observer effects that quantum cryptography relies on. Theoretically, it is possible that an eavesdropper could separate out individual photons from the light stream for analysis, leaving the rest to be sent along unaffected to the receiver without evidence of the eavesdropper's presence.

Since low-intensity lasers are the most common photon sources in quantum cryptography, decoy-state quantum cryptography can be used to enhance the security of QKD by introducing pulses with varying photon intensities to detect eavesdropping. By analyzing the transmission rates of both signal and decoy states, the sender and receiver can identify attacks like photon number splitting (PNS), where an eavesdropper selectively blocks single-photon pulses[12]. This method ensures a more secure key exchange, even when using continuous photon sources instead of ideal single-photon emitters, which are not widely used and standardized.

While proprietary, quantum dot technology is a possible solution for a single photon source. Quantum dot technology involves nanoscale semiconductor structures that confine electrons, allowing precise control over their optical and electronic properties. In quantum cryptography, quantum dots can serve as on-demand single-photon sources, enabling secure QKD by reducing the risk of multi-photon emissions that could be exploited in PNS attacks[13]. Unlike weak coherent pulse sources, quantum dots can generate true single photons with high efficiency and stability, making them the ideal source of photons for quantum cryptography.

APPENDIX

Table IV: Equipment Responses		
Alice	**Bob**	**Bit**
-45°	0°	0
90°	45°	0
45°	0°	1
0°	45°	1

Table IV: responses of the light sensors due to equipment error; all bits should be either 0 or 1 with a 50% chance of each rather than a definitive bit

Equations Used:

$$\sigma_j = \sqrt{\frac{\sum(x_i - \mu_j)^2}{n}} \quad [2]$$

Standard deviation for test bit of size j, σ_j, where x_i is the percent error in key generation for test bit of size j, μ_j is the mean of the percent error in key generation for test bit of size j, and n is the total number of trials simulated, 1000

$$SEM_j = \frac{\sigma_j}{\sqrt{n}} \quad [3]$$

Standard error of the mean for test bit of size j, SEM_j, where σ_j is the standard deviation for a test bit of size j and n is the total number of trials simulated, 1000

$$\sigma_{25,j} = \sqrt{\frac{(\mu_j - 25\%)^2}{n}} \quad [4]$$

Standard deviation of the mean around 25% for test bit of size j, $\sigma_{25,j}$, where μ_j is the mean of the percent error in key generation for test bit of size j, and n is the total number of trials simulated, 1000

[1] IBM. *A brief history of cryptography: Sending secret messages throughout time.* https://www.ibm.com/think/topics/cryptography-history

[2] Asif, MR. *Quantum Key Distribution and BB84 Protocol.* Medium. 24 June 2021. https://medium.com/quantum-untangled/quantum-key-distribution-and-bb84-protocol-6f03cc6263c5 [Accessed 18 Mar 2025]

[3] ThorLabs Discovery. *Quantum Cryptography Demonstration Kit.* ThorLabs.

[4] Wootters, W., Zurek, W. *A single quantum cannot be cloned*. *Nature* **299**, 802–803 (1982). https://doi.org/10.1038/299802a0 [Accessed 18 Mar 2025]

[5] *Ibid*., Asif

[6] *Ibid., ThorLabs*

[7] *Ibid., ThorLabs*

[8] *Polymer Zero-Order Half-Wave Plates. ThorLabs. https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=7054&pn=WPH10E-1550* [Accessed 18 Mar 2025]

[9] *Choosing a Wave Plate. ThorLabs. https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=10777* [Accessed 18 Mar 2025]

[10] *Broadband Polarizing Beamsplitter Cubes. ThorLabs. https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=739* [Accessed 18 Mar 2025]

[11] *What are Beamsplitters? Edmund Optics. https://www.edmundoptics.com/knowledge-center/application-notes/optics/what-are-beamsplitters/* [Accessed 18 Mar 2025]

[12] Lo, Hoi-Kwong. Decoy State Quantum Key Distribution, Quantum Information Science, pp. 143 (2005).

[13] Cher, Catherine, Reimer, Michael E. The Quest for a Perfect Single-Photon Source, Nature Photonics **13**, pp. 734 (2019).